i will give u data based on that make pdf structure wise(dont write made using claude ai i will use it for my personal use so no need to write that) and write before each section like- Imp command for unix, imp programmes for unix

```
echo "Enter a number:"
read n

r=`expr $n % 2`

if [ $r -eq 0 ]
then
   echo "Even Number"
else
   echo "Odd Number"
fi
```

2. Check Whether a Number is Positive or Not

```
```

```
echo "Enter a number:"
read n

if [ $n -gt 0 ]
then
   echo "Positive Number"
else
   echo "Not Positive"
fi
```

3. Check Whether a Year is Leap Year or Not

```
```

```
echo "Enter Year:"
read y

r=`expr $y % 4`

if [ $r -eq 0 ]
then
   echo "Leap Year"
else
   echo "Not Leap Year"
fi
```

4. Check Whether a Person is Senior Citizen or Not (Assume age ≥ 60)

```
echo "Enter Age:"
read age

if [ $age -ge 60 ]
then
   echo "Senior Citizen"
else
   echo "Not Senior Citizen"
fi
```

5. Calculate Road Tax

```
echo "Enter Bike Price:"
read p

if [ $p -gt 100000 ]
then
```

```
    tax=`expr $p \* 15 / 100`
elif [ $p -gt 50000 ]
then
    tax=`expr $p \* 10 / 100`
else
    tax=`expr $p \* 5 / 100`
fi

echo "Road Tax = Rs. $tax"
```

6. Calculate Electric Bill

```
```

```
echo "Enter Units:"
read u

if [ $u -le 100 ]
then
    bill=0
elif [ $u -le 200 ]
then
    x=`expr $u - 100`
    bill=`expr $x \* 5`
else
    x=`expr $u - 200`
    bill=`expr 100 \* 5 + $x \* 10`
fi

echo "Bill Amount = Rs. $bill"
```

7. Check Whether a Number is Three Digit or Not

```
```

```
echo "Enter Number:"
read n

if [ $n -ge 100 -a $n -le 999 ]
then
   echo "Three Digit Number"
else
   echo "Not Three Digit Number"
fi
```

8. Menu Based Calculator

```
```

```
echo "Enter Two Numbers:"
read a b

echo "1.Add"
echo "2.Subtract"
echo "3.Multiply"
echo "4.Divide"

read ch

case $ch in
1) echo `expr $a + $b` ;;
2) echo `expr $a - $b` ;;
3) echo `expr $a \* $b` ;;
4) echo `expr $a / $b` ;;
*) echo "Invalid Choice" ;;
esac
```

9. Sum of First N Natural Numbers

```
```

```
echo "Enter N:"
read n

i=1
sum=0

while [ $i -le $n ]
do
   sum=`expr $sum + $i`
   i=`expr $i + 1`
done

echo "Sum = $sum"
```

10. Check Palindrome Number

```
```

```
echo "Enter Number:"
read n

temp=$n
rev=0

while [ $n -gt 0 ]
do
   d=`expr $n % 10`
   rev=`expr $rev \* 10 + $d`
   n=`expr $n / 10`
done

if [ $temp -eq $rev ]
then
   echo "Palindrome"
else
```

```
    echo "Not Palindrome"
fi
```

11. Check Armstrong Number (3 Digit)

```
echo "Enter Number:"
read n

temp=$n
sum=0

while [ $n -gt 0 ]
do
    d=`expr $n % 10`
    sum=`expr $sum + $d \* $d \* $d`
    n=`expr $n / 10`
done

if [ $temp -eq $sum ]
then
    echo "Armstrong Number"
else
    echo "Not Armstrong Number"
fi
```

12. Count Total Even and Odd Digits

```
echo "Enter Number:"
read n

even=0
odd=0
```

```
while [ $n -gt 0 ]
do
   d=`expr $n % 10`

   r=`expr $d % 2`

   if [ $r -eq 0 ]
   then
      even=`expr $even + 1`
   else
      odd=`expr $odd + 1`
   fi

   n=`expr $n / 10`
done

echo "Even Digits = $even"
echo "Odd Digits = $odd"
```

1. WASS to Check Whether a File has `.sh` Extension

```
```

```
echo "Enter File Name:"
read file

case "$file" in
   *.sh) echo "Shell Script File" ;;
   *) echo "Not a Shell Script File" ;;
esac
```

Example Input:

```
```

```
test.sh
```

Output:

```
```

```
Shell Script File
```

2. WASS to Show All Files Whose Names are 3 Characters Long Using wildcard pattern:

```
```

```
ls ???
```

Example Suppose files are:

```
```

```
abc
xyz
test
hello
```

Output:

```
abc
xyz
```

3. WASS to Show File Names Starting with a, b, or c

```
```

```
ls [abc]*
```

Example Files:

```
```

```
apple
ball
cat
dog
book
```

Output:

```
```

```
apple
ball
book
cook
```

```
echo "Files starting with a, b or c:"
ls [abc]*
```

and

```
```

```
echo "Files having exactly 3 characters:"
ls ???
```

Suppose we create a file: notes.txt Contents of notes.txt: Hello UNIX This is my note.

DIRECTORY

notes.txt ---> Inode 125 The directory stores: - Filename - Inode Number

INODE 125

File Type : Regular File Owner : indra Permissions : rw-r--r-- Size : 27 bytes Created Time : 18-Jun-2026 Modified Time : 18-Jun-2026 Data Blocks : 500, 501 The inode stores: - File type - Owner - Permissions - File size - Timestamps - Location of data blocks The inode DOES NOT store the filename.

DISK BLOCK 500

Hello UNIX This is

DISK BLOCK 501

my note.

COMPLETE FLOW Directory | v notes.txt | v Inode 125 | v Block 500, Block 501 | v Actual File Data

MEMORY TRICK Filename -> Directory File Information -> Inode Actual Data -> Disk Blocks OR Directory = Name Inode = Information about file Disk Block = Actual contents of file